

Hadamard Block Counterexamples to the 1-DPP of ℓ_p

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2026-06-16

Status

This packet records a candidate counterexample, likely valid. It gives a negative answer to a question left open by Karn and Sinha in arXiv:1207.3620: whether ℓ_p , $1 < p < \infty$, has the 1-Dunford–Pettis property.

Source Passage

After proving that ℓ_p does not have the r -DPP for any $r > 1$, and that it does not have several mixed (r, s) -DPP properties with $r > 1$ or $s > 1$, Karn and Sinha leave the endpoint $r = s = 1$ unsettled.

Corollary 3.12 *For $1 < p < \infty$, l_p does not have*

- (a) *The (r, s) -Dunford-Pettis property if $1 < r, s < \infty$.*
- (b) *The $(r, 1)$ -Dunford-Pettis property if $1 < r < \infty$.*
- (c) *The $(1, r)$ -Dunford-Pettis property if $1 < r < \infty$.*

Proof. (a) For $1 < s \leq r$, if l_p has the (r, s) -DPP then it also has the r -DPP. Thus if $1 < s \leq r < \infty$, then l_p does not have the (r, s) -DPP.

Next, let $1 < r < s$. Find $\langle \alpha_n \rangle \in l_s^w(l_p)$ such that $\langle \alpha_n \rangle \notin l_r^w(l_p)$. Find $\beta \in l_{p'}$ such that $\sum_{n=1}^{\infty} |\langle \beta, \alpha_n \rangle|^r = \infty$. Putting $\beta_1 = \beta$ and $\beta_n = 0$ for $n \geq 2$, $\langle \beta_n \rangle \in l_r^w(l_{p'})$. However,

$$\langle \langle \langle \beta_k, \alpha_n \rangle \rangle_k \rangle_n \notin l_s^s(l_r).$$

Thus l_p does not have the (r, s) -DPP for $1 < r, s < \infty$.

Now, both (b) and (c) can be obtained on the lines of (a). $\triangle$

Note We have not been able to settle whether for $1 < p < \infty$, l_p has the 1-DPP.

Transcription of the Question

The note on page 11 states:

“We have not been able to settle whether for $1 < p < \infty$, ℓ_p has the 1-DPP.”

Definitions Used

We use the characterization from Karn and Sinha’s Theorem 3.7 [1]: a Banach space X has the 1-Dunford–Pettis property if and only if

$$\Pi_1(X, \ell_1) = \mathcal{B}(X, \ell_1).$$

Thus, to disprove the 1-DPP for $X = \ell_p$, it is enough to construct a bounded operator $T : \ell_p \rightarrow \ell_1$ which is not absolutely 1-summing.

For a bounded operator $S : X \rightarrow Y$, the absolutely 1-summing norm is denoted by $\pi_1(S)$. We use the standard finite-sequence lower test

$$\pi_1(S) \geq \frac{\sum_i \|Sx_i\|}{\sup_{\varphi \in B_{X^*}} \sum_i |\varphi(x_i)|}.$$

The Counterexample

Theorem 1. *For every $1 < p < \infty$, the space ℓ_p does not have the 1-Dunford–Pettis property.*

Proof Intuition

The obstruction is finite-dimensional. A Hadamard matrix H_N , viewed as an operator $\ell_p^N \rightarrow \ell_1^N$, has small operator norm because $\|H_N x\|_1 \leq N\|x\|_2$. But its absolutely 1-summing norm is large: on the N unit vectors, all columns have ℓ_1 -norm N , while the weak ℓ_1 -size of the unit vectors in ℓ_p^N is only $N^{1/p}$. Therefore $\pi_1(H_N)/\|H_N\| \rightarrow \infty$.

The infinite counterexample is a block diagonal sum of these finite matrices. The block scalars make the operator norms summable in the correct $\ell_{p'}$ -sense, so the resulting operator $\ell_p \rightarrow \ell_1$ is bounded. At the same time, the 1-summing norm on the individual blocks still tends to infinity. A 1-summing operator cannot have restrictions with unbounded 1-summing norm, so the block sum is bounded but not absolutely 1-summing.

Proof. Let p' be the conjugate exponent of p . For each $N = 2^j$, let H_N be the Sylvester Hadamard matrix, so its entries are ± 1 and

$$H_N^t H_N = N I_N.$$

Regard H_N as an operator $\ell_p^N \rightarrow \ell_1^N$.

First we estimate its operator norm. For every $x \in \mathbb{R}^N$ or $\mathbb{C}^N$,

$$\|H_N x\|_1 \leq N^{1/2} \|H_N x\|_2 = N \|x\|_2.$$

Therefore

$$\|H_N : \ell_p^N \rightarrow \ell_1^N\| \leq B_N := \begin{cases} N, & 1 < p \leq 2, \\ N^{3/2-1/p}, & 2 \leq p < \infty, \end{cases}$$

using $\|x\|_2 \leq \|x\|_p$ for $p \leq 2$ and $\|x\|_2 \leq N^{1/2-1/p}\|x\|_p$ for $p \geq 2$.

Next we lower-bound $\pi_1(H_N)$. Test on the unit vectors $e_1, \dots, e_N$ of ℓ_p^N . Each column $H_N e_i$ has ℓ_1 -norm N , and

$$\sup_{\|a\|_{p'} \leq 1} \sum_{i=1}^N |\langle a, e_i \rangle| = \sup_{\|a\|_{p'} \leq 1} \|a\|_1 = N^{1/p}.$$

Hence

$$\pi_1(H_N) \geq \frac{N^2}{N^{1/p}} = N^{2-1/p}.$$

It follows that

$$\frac{\pi_1(H_N)}{B_N} \geq R_N := \begin{cases} N^{1-1/p} = N^{1/p'}, & 1 < p \leq 2, \\ N^{1/2}, & 2 \leq p < \infty. \end{cases}$$

In either case $R_N \rightarrow \infty$ along $N = 2^j$.

Now take $N_j = 2^j$, $R_j = R_{N_j}$, and $B_j = B_{N_j}$. Set

$$c_j = R_j^{-1/2}, \quad A_j = \frac{c_j}{B_j} H_{N_j} : \ell_p^{N_j} \rightarrow \ell_1^{N_j}.$$

Then $\|A_j\| \leq c_j$. Since R_j grows exponentially, the scalar sequence (c_j) belongs to $\ell_{p'}$. Identify

$$\ell_p \cong \left(\bigoplus_{j=1}^{\infty} \ell_p^{N_j} \right)_{\ell_p}, \quad \ell_1 \cong \left(\bigoplus_{j=1}^{\infty} \ell_1^{N_j} \right)_{\ell_1}.$$

Define the block diagonal operator

$$T((x_j)_{j=1}^{\infty}) = (A_j x_j)_{j=1}^{\infty}.$$

For $x = (x_j) \in \ell_p$, Holder's inequality gives

$$\|Tx\|_1 = \sum_j \|A_j x_j\|_1 \leq \sum_j c_j \|x_j\|_p \leq \|(c_j)\|_{p'} \left(\sum_j \|x_j\|_p^p \right)^{1/p}.$$

Thus $T : \ell_p \rightarrow \ell_1$ is bounded.

However,

$$\pi_1(A_j) = \frac{c_j}{B_j} \pi_1(H_{N_j}) \geq c_j R_j = R_j^{1/2} \longrightarrow \infty.$$

If T were absolutely 1-summing, then for the coordinate inclusion $J_j : \ell_p^{N_j} \rightarrow \ell_p$ and coordinate projection $P_j : \ell_1 \rightarrow \ell_1^{N_j}$, the ideal property would imply

$$\pi_1(A_j) = \pi_1(P_j T J_j) \leq \pi_1(T)$$

for every j , a contradiction. Therefore T is bounded but not absolutely 1-summing.

By Karn and Sinha's characterization $\Pi_1(\ell_p, \ell_1) = \mathcal{B}(\ell_p, \ell_1)$ if and only if ℓ_p has the 1-DPP. The operator T shows this equality fails, and hence ℓ_p does not have the 1-DPP. $\square$

Verification Notes

The proof is analytic. The included script `code/verify_hadamard_bounds.py` verifies the finite Hadamard identities $H_N^t H_N = N I_N$ and the column ℓ_1 -norms for small Sylvester matrices, then prints the lower and upper estimates used above for sample values of p . It is a sanity check, not a substitute for the proof.

Novelty and Literature Check

The run indexes were searched for arXiv:1207.3620, the paper title, “operator summability”, “operator p -summability”, “1-DPP”, “ $\Pi_1(\ell_p, \ell_1)$ ”, and close variants. No prior packet or attempt in the run addressed this endpoint question.

The local parsed arXiv corpus was searched for later citations to Karn–Sinha and for exact phrases around 1-DPP, ℓ_p , and absolutely 1-summing operators into ℓ_1 . Several related papers cite the source, but this bounded pass found no explicit later answer to the note. External web searches for exact phrases around “ ℓ_p 1-DPP”, “ ℓ_p 1-DPP An operator summability”, “ $\Pi_1 \ell_p \ell_1$ 1-summing”, and “absolutely 1-summing $\ell_p \ell_1$ ” likewise surfaced no explicit later resolution. Novelty confidence is moderate: the result appears new to the run memory and to this bounded search, but the literature check is not exhaustive.

Limitations

This answers the endpoint question for the classical spaces ℓ_p , $1 < p < \infty$. It does not attempt to classify all spaces for which every operator into ℓ_1 is absolutely 1-summing, nor does it address the separate Section 4 question in the source paper about ℓ_p^t -completeness of the operator ideal $Lt_p^*(X, Y)$.

Human Review Recommendation

Reviewers should check three points first: the finite-dimensional $\pi_1(H_N)$ lower bound obtained from the unit vectors, the two-case operator-norm estimate through ℓ_2 , and the weighted block-diagonal argument showing boundedness while the block restrictions have π_1 -norms tending to infinity.

References

- [1] A. K. Karn and D. P. Sinha, *An operator summability of sequences in Banach spaces*, Glasgow Math. J. **56** (2014), 427–437; arXiv:1207.3620.
- [2] J. Diestel, H. Jarchow, and A. Tonge, *Absolutely Summing Operators*, Cambridge Studies in Advanced Mathematics 43, Cambridge University Press, 1995.